25900689 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEZGYA01 - FUNDAMENDALS OF SOIL ECOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Question

Answer all questions

Each question carries **1** marks

1. Actinorhizal plants are typically associated with: [K] / [CO1, CO2, CO3, CO4, CO5]

   a). Legumes b). Non-leguminous woody plants

   c). Cereals d). Algae

2. Mineralization is the process of: [K] / [CO2]

   a). Formation of rocks b). Conversion of organic matter in soil into inorganic nutrients

   c). Water evaporation d). Root elongation

3. Biological interactions in soils mainly include: [K] / [CO3]

   a). Predation, competition, mutualism, parasitism b). Respiration and photosynthesis

   c). Carbon fixation only d). Evaporation and precipitation

4. Loamy soil is considered ideal for crops because: [U] / [CO5]

a). It is rich in organic matter only

b). It has balanced proportions of sand, silt, and clay

c). It is always dry

d). It has only micropores

5. Why are nematodes considered particularly indicative organisms for assessing the status of the soil? [K] / [CO1, CO2, CO3, CO4, CO5]

a). They are not very diverse, making them easily characterized

b). They are present in high numbers, with high species diversity and provide good information on the diversity and stability of the soil ecosystem

c). They are scarce

d). They are primarily used by ecotoxicologists to determine the presence of pollutants

6. Actinorhiza relationships are associated with fixation of: [U] / [CO1, CO2, CO3, CO4, CO5]

a). Oxygen

b). Nitrogen

c). Phosphorus

d). Sulfur

7. The litterbag method is used to measure: [K] / [CO2]

a). Root growth

b). Litter decomposition

c). Microbial respiration

d). Soil porosity

8. Anaerobic decomposition produces ___. [U] / [CO2]

a). CH_4

b). O_2

c). NO_3

d). NH_4

9. Which fraction of soil organic matter is described as the "final product of decomposition" and contributes to soil structure and darkening of soil color? [K] / [CO1, CO2, CO3, CO4]

a). Plant residues

b). Active soil organic matter

c). Humus

d). Living microbial biomass

10. The microbial conversion of organic nitrogen into ammonium is called ___. [U] / [CO2]

a). Ammonification b). Nitrification

c). Denitrification d). Assimilation

[1x10 = 10]

Part B

Short Answer

Answer any **14** questions

Each question carries **2** marks

11. What are soil horizons? [U] / [CO5]

12. Illustrate the link between decomposition and nutrient cycling? [U] / [CO2]

13. Explain how soils contribute to the global carbon cycle. [U] / [CO3]

14. Define megafauna and list two examples of megafauna. [K] / [CO3]

15. Explain the role of soil microorganisms in humus formation. [U] / [CO2]

16. Why are soil microbes called “ecosystem engineers”? [U] / [CO1, CO2, CO3, CO4, CO5]

17. Why is soil considered "dynamic"? [K] / [CO1, CO2, CO3, CO4, CO5]

18. What is a common initial step when isolating microorganisms from a soil sample? [K] / [CO5]

19. Discuss why mesofauna are important indicators of soil health. [U] / [CO3]

20. Compare and contrast the feeding strategies of macrofauna detritivores (earthworms, millipedes) versus mesofauna detritivores (springtails, mites). [U] / [CO2]

21. How do root exudates influence soil microbial activity? [U] / [CO1, CO2, CO3, CO4, CO5]

22. How does lignin content affect the decomposition of plant materials? [U] / [CO2]

23. State the ecological role of root zone communities. [K] / [CO3]

24. Describe the structural difference between actinorhizal nodules and rhizobial nodules. [K] / [CO1, CO2, CO3, CO4, CO5]

25. Give an example of a soil insect and its ecological function. [U] / [CO3]

26. Explain the role of mycorrhizal fungi in forest decomposition processes. [U] / [CO2]

[2x14 = 28]

Part C

Short Essay

Answer any **8** questions

Each question carries **4** marks

27. Summarize the dynamics of litter breakdown. [U] / [CO2]

28. Write a note on symbiotic associations of roots with microbes. [U] / [CO1, CO2, CO3, CO4, CO5]

29. Describe characteristics of any five soil types. [U] / [CO5]

30. How does soil type affect decomposition? [U] / [CO2]

31. Define field capacity in soils. [K] / [CO5]

32. Write a note on the role of protists in nutrient cycling. [U] / [CO1, CO2, CO3, CO4, CO5]

33. Explain the ecological role of nematodes and tardigrades in soil ecosystems. [U] / [CO3]

34. How do fauna influence microbial activity indirectly? [U] / [CO2]

35. Explain the process of primary production in soils with reference to energy capture. [U] / [CO1, CO2, CO3, CO4, CO5]

36. Compare colonial growth forms and mycelial growth forms in terms of structure and function [U] / [CO3]

[4x8 = 32]